



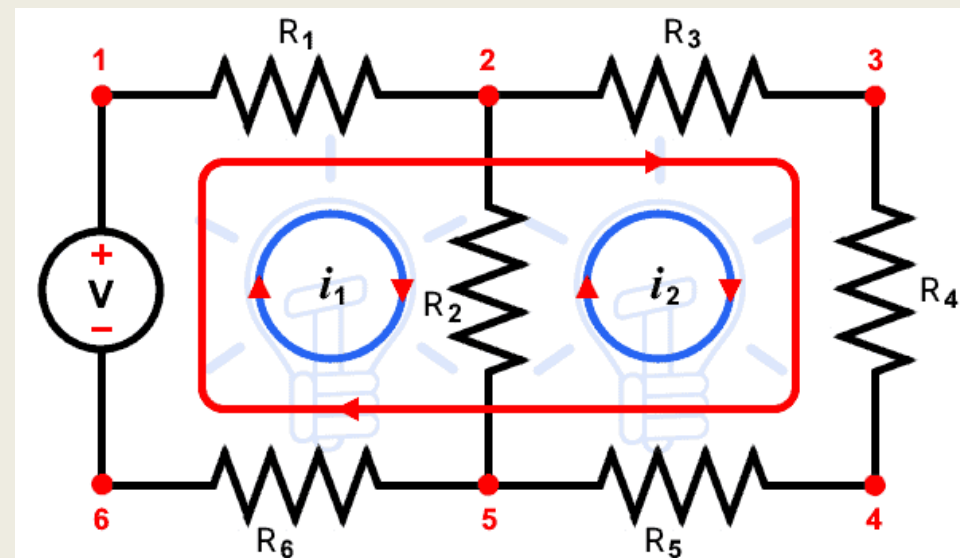
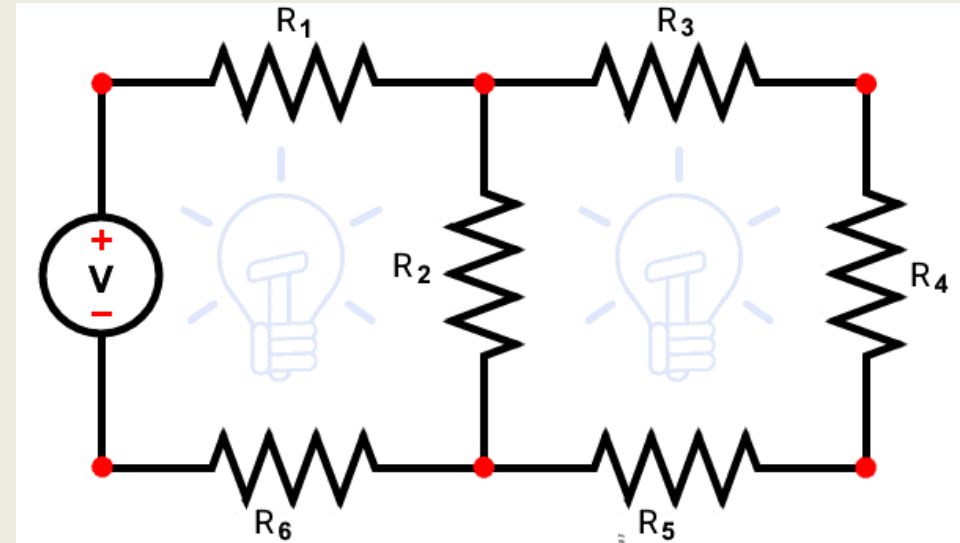
LOOPS AND BRANCHES

Concept Clearance

Loop

Any individual closed path in a circuit. The starting node and ending node of a path is same. In the shown figure it exists three loops.

Note: সার্কিটকে সমতলে আঁকতে হবে। দি-মাত্রিক সারফেসে হবে, ত্রি-মাত্রিক হলে হবে না। ত্রি-মাত্রিক থাকলে সেটাকে দি-মাত্রিক বানিয়ে নিতে হবে।

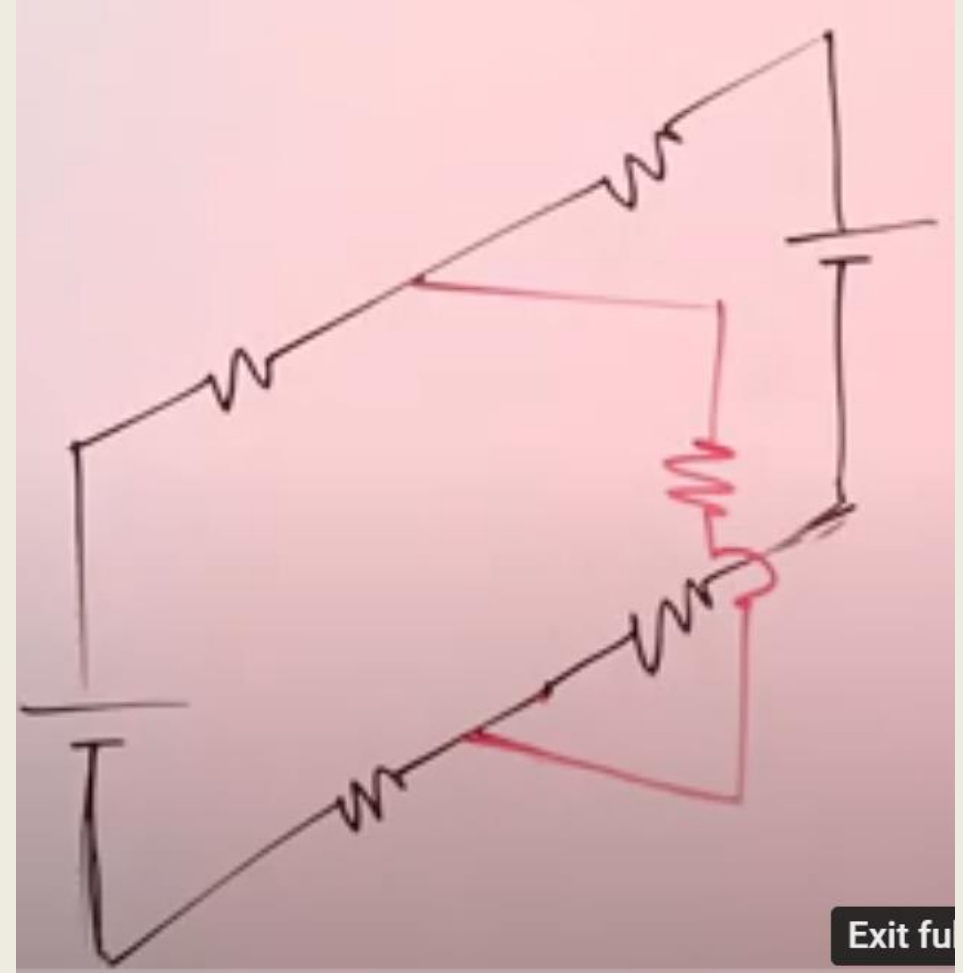


6 Nodes, 7 Branches, 3 Loops & 2 Meshes

Loop

Any individual closed path in a circuit. The starting node and ending node of a path is same.
In the shown figure it exists three loops.

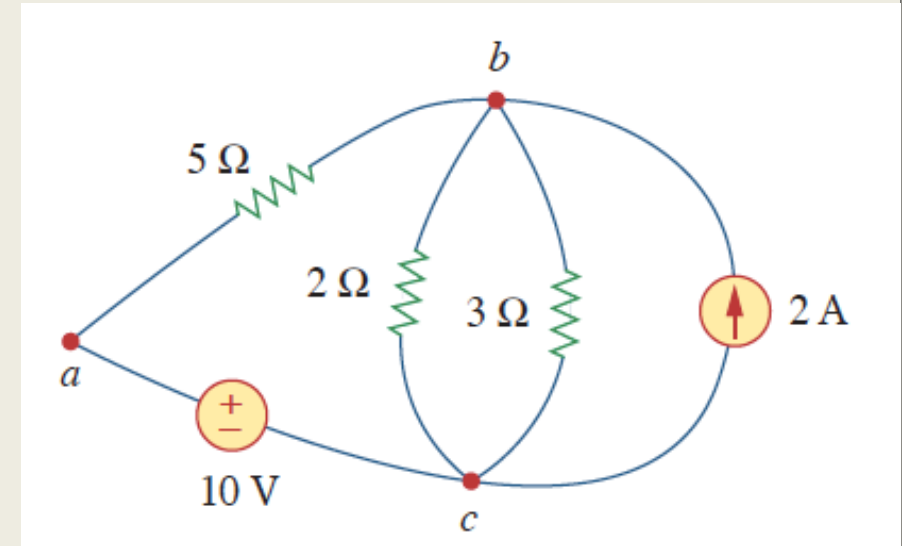
Note: সার্কিটকে সমতলে আঁকতে হবে। দি-মাত্রিক সারফেসে হবে, ত্রি-মাত্রিক হলে হবে না। ত্রি-মাত্রিক থাকলে সেটাকে দি-মাত্রিক বানিয়ে নিতে হবে।



Dependent and Independent Loop

Independent Loop: If elements of each loops are unique then the loops are called independent loop.

- In the figure shown in the right side, it has 6 loops. Among them three meshes and two independent loops.



Mesh

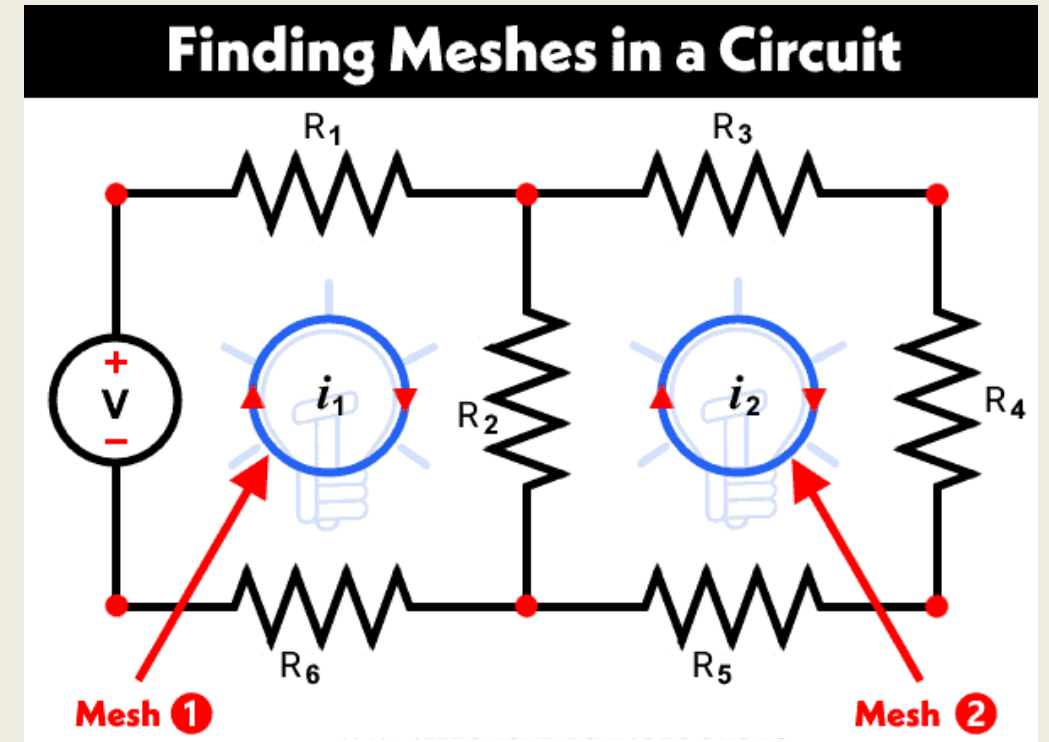
If two loops does not overlap each other, including all of the circuit elements. then the loops are called mesh.

In the aforementioned figure, loop3 is overlapped. Thus it is not a mesh. In the figure shown in right side, two meshes are indicated.

You may say that, I can mention loop3 as an individual node neglecting loop1 and loop2. But in this case Resistor R_2 will be neglected.

So we have two rules of mesh:

1. No element will be neglected.
2. There must be non-overlapping loops.



Mesh

We need to have clear concept of mesh to solve mesh analysis that we will learn later.